HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **schema changes**, including **exact numeric types** and **audit trails**, for new features within a **PostgreSQL** database supporting a regulated derivatives exchange.
- Wrote and reviewed **lock-safe and reversible database migrations** for large tables, ensuring data integrity and minimal downtime during production deployments.
- Analyzed **execution plans** for slow queries identified in production metrics, optimizing **PostgreSQL** performance by adjusting queries, indexes, and access patterns.
- Developed **Python scripts** to automate data extraction and transformation tasks for financial reporting, reducing manual processing time by approximately **15 hours per week**.
- Managed **production incidents** for enterprise financial systems, performing root cause analysis and implementing fixes within strict **SLA** targets.
- Collaborated with cross-functional teams to define requirements and support User Acceptance Testing (UAT) for system integrations and configuration updates.
- Containerized a Java Spring Boot application using **Docker** and deployed it to an **AWS EC2** instance, ensuring consistent environments for development and production.

SKILLS

Languages – Haskell, SQL, Python, Java, C++, Rust

Databases – PostgreSQL, MySQL, Oracle

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

Tools & Technologies – Docker, Jenkins, Git, Jira, Confluence, Splunk, Dynatrace, PagerDuty, Grafana, Datadog, Atlas

Concepts – Distributed Systems, Concurrency, Replication, Consistency, Idempotency, Transactions, Isolation Levels, Schema Design, Database Migrations, Order Flow, Trade Flow, Clearing, Settlement, Audit Trails, Financial Messaging

PROJECTS

Exchange Data Model and Migration Service

- Designed the relational schema for order, trade, and settlement data in **PostgreSQL**, incorporating **exact numeric types** and comprehensive **audit trails** to prevent financial value drift.
- Developed a suite of **lock-safe, reversible migrations** for high-volume tables using SQL, facilitating seamless deployments and rollbacks for the derivatives exchange platform.
- Implemented **transactional integrity** across critical data flows, ensuring atomicity for order placements and trade executions within the core exchange services.

Financial Operations Data Pipeline Automation

- Built automated **Python ETL pipelines** to ingest and process transaction data from multiple sources, supporting reconciliation workflows for a financial services client.
- Developed **SQL queries** and scripts to validate data integrity and generate scheduled reports on payment processing metrics, reducing manual effort by approximately **30%**.
- Integrated the data pipeline with **AWS S3** for storage and archival, improving data accessibility for downstream analytics and operational monitoring.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)